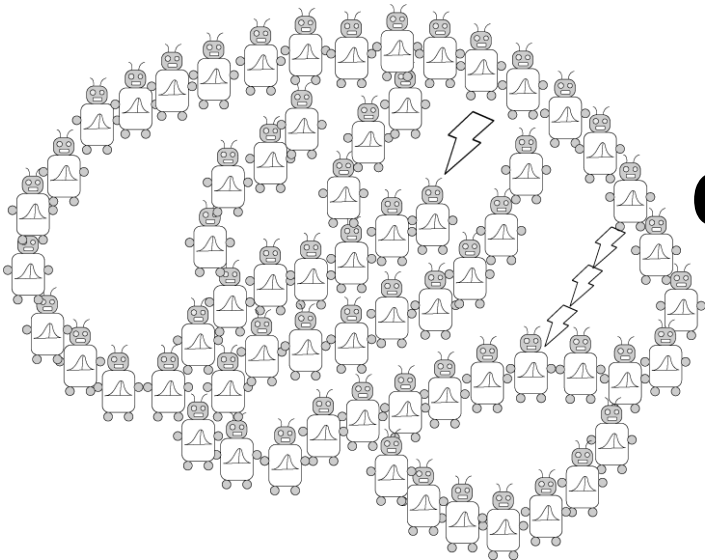




Introduction



**PSYCH/COMPSCI/BMI 841:
Computational Cognitive Science**

Joseph Austerweil

January 23, 2025

Deductive and inductive

The problems that people excel relative to machines are inductive problems.

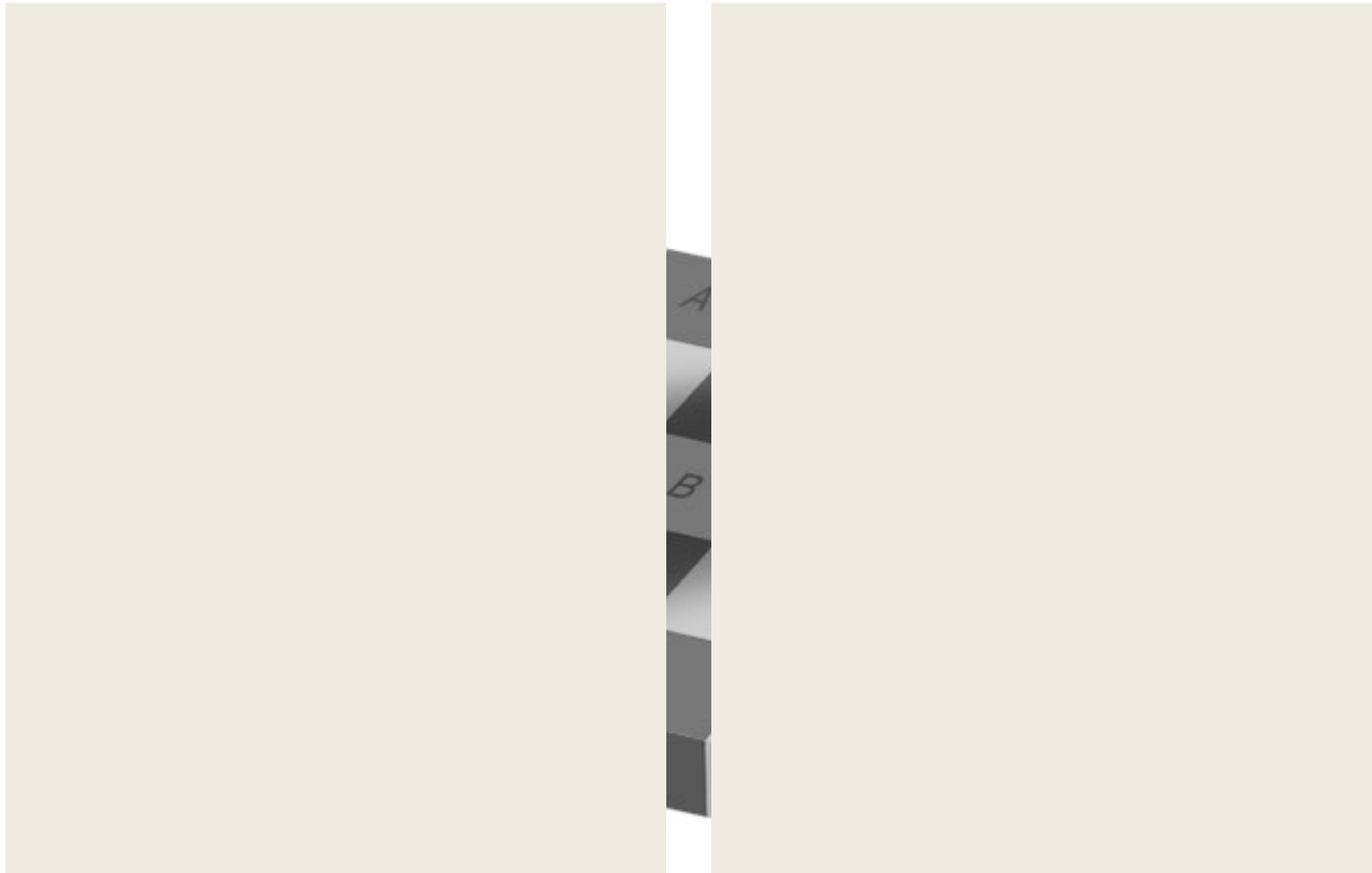
They feel easy to solve, but REALLY HARD!

Inductive problems involves going from observed data to underdetermined answers.

Requires knowledge of the rules of the system, but also expectations about possible answers



Visual illusions



http://web.mit.edu/persci/people/adelson/checkershadow_illusion.html

Perception and induction

$$? + ? = 5$$

square color + shadow = intensity

Your visual system solves this inductive problem, using rules about how retinal images are generated, and some expectations about color and shadow

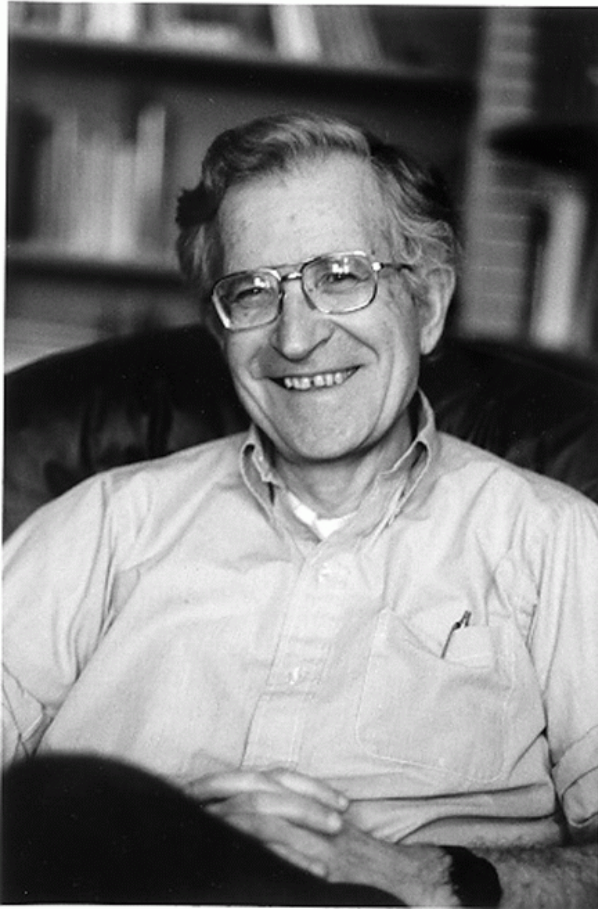
Learning language

data: utterances produced by people

hypotheses: all possible ways of combining sounds



The poverty of the stimulus



Noam Chomsky

Utterances do not contain enough information to determine a language.

What are the rules of the system, and expectations about answers that humans have?

Speech recognition



hypotheses: speech sounds
 accent of speaker
 segmentation into words

How can human children go from raw sound to the structured representations required to understand and learn language?



Interpreting sentences

data: “The woman hit the man with the umbrella.”

hypotheses: what actually happened?

Interpreting sentences

“The woman hit the man with the hammer.”

data: “The woman hit the man with the umbrella.”

“The woman hit the man with the mustache.”

hypotheses: what actually happened?

How do people integrate expectations about the world
with syntactic rules to interpret sentences?

Inferring mental states

<https://www.youtube.com/watch?v=VTNmLt7QX8E>

(Heider & Simmel, 1958)

Inferring mental states

data: behavior of others

hypotheses: goals and beliefs

Are there rules that describe how we expect goals and beliefs to relate to behavior?

Expectations we have about the goals and beliefs of others?

Learning new concepts

“tufa”



“tufa”

“tufa”

Learning new concepts

data: examples

hypotheses: possible sets of objects,
or ways of describing those sets

Are there rules that describe how we expect examples to relate to concepts?

What are the expectations we have about good concepts?



Learning words



“jumbuck”

Scientific Discovery and Causal Learning

“The contagion spread rapidly and before its progress could be arrested, sixteen persons were affected of which two died. Of these sixteen, eight were under my care. On this occasion I used for the first time the affusion of cold water in the manner described by Dr. Wright. It was first tried in two cases ... The effects corresponded exactly with those mentioned by him to have occurred in his own case and thus encouraged the remedy was employed in five other cases. It was repeated daily, and of these seven patients, the whole recovered.”



Currie (1798)
Medical Reports on, the Effects of Water, Cold and Warm, as a Remedy in Fevers and Febrile Diseases

Scientific Discovery and Causal Learning

	<i>Treated</i>	<i>Untreated</i>
<i>Recover</i>	7	7
<i>Die</i>	0	2

“Does the treatment cause recovery?”

Inferring causal relations

data: interaction between objects

hypotheses: existence of causal relation

Are there rules that describe how we expect causes to relate to effects?

When do we think causal relations likely, and what properties do we expect?

The big questions

1. How does knowledge guide learning, inference, and decision-making from sparse, noisy or ambiguous data?
2. What are the form and contents of this knowledge?
3. How is this knowledge itself learned?
4. How do we balance constraint and flexibility, assimilating new data to our current model versus accommodating our model to the new data?
5. How can accurate inductive inferences be made efficiently, even with complex hypothesis spaces?

A strategy: Reverse engineering

Specify computational problems

Identify ideal solutions to these problems

Examine how well these solutions account for human behavior.

See how well they work for machine learning.

Outline for rest of class

Administrative Matters

Short Interlude

Basic Bayes

Now administrative matters...

Office hours

Joe: Tu 2-3pm

zoom via Canvas

Also by appointment

(in-person possible)

Life update

Email policy

I will do my best to respond to emails within 36 hours (48 hours if you send it late Friday or over the weekend).

Please do your best to keep them clear and concise.

Course topics

Cover parallel developments in human and machine learning

We will primarily focus on probabilistic models, but we will spend a few weeks on deep learning too.

(your project does not have to – more on that later).

Topics:

- Graphical models/Bayesian networks
- Hierarchical Bayesian models
- Markov chains/Foraging
- Reinforcement learning/Social reasoning/Communication
- Approximation: [Markov chain] Monte Carlo
- Bayesian nonparametrics
- Ethics/Prejudice in ML
- Deep learning and hybrid models

Course learning goals

Students who successfully complete the class will be able to

1. **understand** state-of-the-art machine learning techniques.
2. **create** a probabilistic model of a problem solved by the human mind (hopefully one relevant to your research interests).
3. **analyze** how model assumptions (explicit or implicit) enable learning.
4. **think critically** about the role of computational modeling in explaining human behavior.



Prerequisites

Prior experience with computational modeling and computer programming is required.

Most supported: Python

Supported: R

Legacy very supported: Matlab

Next support (in-progress): Julia

Other languages: I have coded many languages and likely will be able to help, but I can't guarantee it.

Prior familiarity with linear algebra and probability theory is highly recommended.



Typical Class Flow

This and the next two weeks will be atypical.

Today's class is mostly Joe lecturing.

If time, we will do class introductions.

Weeks after that will be typical:

1. First half of class will be Joe lecturing.
2. Second half will be two student presentations and discussion.

Next week will be closer to typical. Joe will lecture for most of class, but will have more discussion.

Third week will be very atypical because Joe will be in Japan.
It will be completely async.

Optional Sections?

To be discussed with class for interest

People have very different backgrounds in the class. To avoid as many bored stares, the purpose of this optional section is reviewing math that went fast during class.
going through programming tutorials together.
etc

I answer questions that you have about math and programming.

They evolve into extra office hours often.

Some were recorded already and accessible to watch.
Rest of lecture today will have some of this.

Typical Weekly Rhythm

	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Reading	Article Reading						
Discussion Post Writing			Write and submit discussion post by 8:59pm				
Lecture and Student Presentations					1230-3pm PSYC 138		
Quiz (if there is one)		Take quiz reviewing last week's material by Wed at 8:59pm					
Assignment if there is one				Due at 8:59pm			
Project/Assignment work	Work towards completion					Work towards completion	

Grading/Requirements

No required. Recs:

If you are concerned about catching up on probabilistic modeling, this book is a good one to look at:

Kruschke (2016). Doing Bayesian data analysis

If you want a book as reference, two good ones are:

Murphy (2012) Machine Learning: A Probabilistic Perspective

Goodfellow et al. (2016): Deep Learning

Requirement	Percentage of final grade
Discussion posts	5% (5x1% each)
Assignments	16% (1&2=4%, 3=6%, & 4 = 2%)
In-class discussion	5%
In-class presentations	10%
Project proposal	10%
Final project presentation	7.5%
Final project proposal	30%
“Weekly quizzes”	16.5%

Discussion Posts (5% total = 1% each)

You are expected to submit a (50-200 word) **reaction** to the reading for the week on the discussion board.

NOT A SUMMARY.

Should focus on one paper.

These are due by 8:59pm on Tuesday of each week.

The first one is due on 02/03 and last one is due on 04/21.

You are only required to do 5 of the 11 posts.

They are pass/fail and submitted through Canvas.



Assignments (16% total)

4 shortish mathematics/coding assignments

Submit through Canvas

Goal to solidify knowledge of computational topics from class.

As a course, I will mainly support Python and R:

Legacy Matlab support is available

I'm hoping to add Julia to the list.

Resources for them have been posted on Canvas.

You can use other programming languages, and I will still try to help. But, I might not be able to.

#3 is harder and longer than rest, so it's worth 1.5x.

#4 is on RL, which has more helping code than the others.

Class presentations (2x 5% each)

Present the major results of one of the course readings.

Must be a non-optional reading.

Also, have 4-6 discussion questions at the end.

These should guide discussion.

Being controversial/provocative is encouraged!

Have to give two if you are taking the course for credit.

One if formally auditing.

Each presentation (including discussion) should be no longer than 25 minutes.

You are encouraged to meet with me beforehand.

Must submit to me on Canvas right before or after (8:59pm day of presentation deadline)!

Class presentations (2x 5% each)

Grade is determined by the following factors:

Understanding of the paper	(2 pts)
Covering the key aspects of the paper	(1 pt)
Presentation clarity	(1 pt)
Appropriate discussion questions	(0.5 pts)
Appropriate use of time	(0.5 pts)
Not using Prezi	(10 ⁹⁹⁹ deduction and have to clean up)

See guidelines and tips document posted online.



Class presentations (2x 5% each)

Focus on the following questions:

1. What is the question that this paper explores? If applicable, what aspect of human behavior are we trying to understand?
2. Any background/context for the paper? Previous work?
3. What is the underlying computational problem being solved? How does the paper formulate it?
4. What is the solution to the computational problem?
(Go for intuitions more than rigor.)
5. How is the solution used to answer the question? How does it compare to other approaches/human behavior?
6. What conclusions can we draw from the results?
What is your interpretation?



Class project (47.5% total)

Opportunity for you to explore a question related to human and machine learning that YOU are interested in.

Different people have different backgrounds, and you should draw on your strengths and interests.

You do not not have to use a topic or technique from class.
(non-exclusive) Categories of topics:

Experiment: Conduct an experiment that tests a prediction of a model from class (or simple extension).

Modeling/ML: Take some human behavior (results from class or scraped data from the Internet) and model it using your own or an advanced ML method.

Math: Math analysis of learning problems/models.



Project proposal (due 03/12 at 8:59pm)

1 page (single-spaced) with four sections:

Background. Identify the topic of your project and briefly provide some context to the project. Summarize any previous research on the topic.

Question. State the question examined by your project. This should be 1-2 sentences (possible points off for more)

Method. Briefly describe the method you are going to use. Give some of the concrete details behind your experiment procedure, modeling approach, data analysis, or math.

References. Have at least three references cited. Only one reference can be a paper from class.

Worth 10 points; Pass/Fail



Project presentations (in-class 05/01)

Goal is to give everyone an opportunity to see the awesome things everyone has done for their projects!

Not expected to have finished your project, but have some preliminary results (beyond the proposal).

At most 10 minutes long.*

Aim to communicate four things:

Background. Previous work to understand your Q.

Question. The question you examine.

Method. The method you used (or are going to use)

Results. Preliminary results you have obtained.

Worth 7.5 points

Submit on Canvas by 8:59pm 05/01.



Final paper (due 05/06 at 8:59pm)

~6 pages single-spaced.

No strict guidelines, but recommendations on Canvas.

Most important factor is to motivate and present your results in a clear and succinct manner.

More details later on in class.

Submit through Canvas.

Worth 30 points.



Lateness policies:

Discussion posts: No late posts. Post a different week.

Assignments: Three “free” days which you get to use throughout the semester. Can be distributed in any manner. Not prorated (12 hours late -> 1 day late; 25 hours -> 2 days). Afterwards, 20% off assignment per day late (not prorated).

Proposal and paper: No lateness allowed (for proposal). Again, 20% off assignment per day late (not prorated).

Class recording policies:

Lectures by professor will be recorded.

I will upload them to Canvas shortly after class.

Paper presentations and discussions will not be recorded.



Plagiarism policies:

All submitted work will be analyzed for plagiarism.

Don't do it.

Note that self-plagiarism (using your own writing from previous classes or trying to get “double” credit) is still plagiarism unless I give you explicit permission.

Plagiarism will not be tolerated and each case will be dealt with on an individual basis.

Generative AI Policy

- Overall, welcome to use tools as a resource for facilitating the course learning objectives.
- For technical issues, they have become an indispensable tool and banning their use would be counterproductive from both pedagogical and research standpoints.
- However, **you are prohibited from asking quiz or assignment questions directly to them.**
- **Your answers to quizzes and assignments can be informed by AI tools, but the majority of an answer should be written in your own words.**
- **Use must be cited.**
- **Rest of class, you are expected to use the following policy:**
 - Your use of AI tools must be documented and cited to conform to this course's expectations.
 - **Please be aware you are responsible for how the information is used when you submit information based on an AI query**

A short interlude...

